GRADE 12 / Life Sciences / English

Monohybrid inheritance

01 Understand

Learning goals: Distinguish genotype and phenotype • Construct a Punnett square • Interpret probability without promising outcomes

Alleles are alternative forms of a gene. In a simple complete-dominance model, one dominant allele is enough for the dominant phenotype.

Genotype describes the allele combination; phenotype is the observable characteristic. A heterozygote has two different alleles.

Each fertilisation is a new probability event. Expected ratios describe many offspring, not a guaranteed sequence within one family.

Tt × Tt: each parent forms T and t gametes

Gamete	T	t
T	TT	Tt
t	Tt	tt

02 Follow the reasoning

In a hypothetical plant, T is dominant for tall growth and t is recessive for short growth. Cross two heterozygotes.

1. Parents: Tt × Tt. Each produces T and t gametes.

2. The combinations are TT, Tt, Tt and tt: genotype ratio 1 : 2 : 1.

3. Tall offspring probability = 3/4; short = 1/4. Phenotype ratio is 3 tall : 1 short under the stated model.

GRADE 12 / Life Sciences

Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Cross Tt with tt and list the offspring genotypes.

2. State the probability of short offspring in that cross.

3. Explain why two tall parents can produce a short offspring.

04 Check and correct

1. Tt, tt, Tt, tt: 1 Tt : 1 tt.

2. 1/2, or 50%.

3. Both may be heterozygous and each may contribute a recessive t allele, producing tt.

Common mistake to avoid

Dominant does not mean more common, stronger or better. State the inheritance model before interpreting a ratio.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Genetics and inheritance: monohybrid crosses

DBE CAPS, Grades 10–12: Life Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID lisc-12.